National University of Computer and Emerging Sciences
Lahore Campus

Operating Systems (CS2018)

Date: May 20, 2024

Course Instructor(s)

Mr. Kamran Jamal

Final Exam

Total Time (Hrs):	3
Total Marks:	80
Total Questions:	4

121-7664 Roll No

6A Section

Student Signature

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Attempt all the questions.

CLO 2 Compare Elements of Processes and Threads

Q1: Five batch jobs. *A* through *E*, arrive at a computer center at almost the same time. They have estimated running times of 10, 6, 2, 4, and 8 minutes. Their (externally determined) priorities are 3, 5, 2, 1, and 4, respectively, with 5 being the highest priority. For each of the following scheduling algorithms, determine the mean process turnaround time. Ignore process switching overhead. **[20]**

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(a) Round robin.
(b) Priority scheduling
(c) First-come, first-served (run in order 10, 6, 2, 4, 8).
(d) Shortest job first.

For (a), assume that the system is multiprogrammed, and that each job gets its fair share of the CPU. For (b) through (d), assume that only one job at a time runs, until it finishes. All jobs are completely CPU bound.

CLO 3 Analyze Memory Management Techniques

Q2: **(a)** A computer has four page frames. The time of loading, time of last access, and the *R* and *M* bits for each page are as shown below (the times are in clock ticks): **[10]**

Page	Loaded	Last ref.	R	M
0	126	280	1	0
1	230	265	0	1
2	140	270	0	0
3	110	285	1	1

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(i) Which page will NRU replace?
(ii) Which page will FIFO replace?
(iii) Which page will LRU replace?
(iv) Which page will second chance replace?

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(b) A computer provides each process with 65,536 bytes of address space divided into pages of 4096 bytes each. A particular program has a text size of 32,768 bytes, a data size of 16,386 bytes, and a stack size of 15,870 bytes. Will this program fit in the machine's address space? Suppose that instead of 4096 bytes, the page size were 512 bytes, would it then fit? Each page must contain either text, data, or stack, not a mixture of two or three of them. **[10]**

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CLO 4 Appraise Characteristics of File Systems

Q3: (a) Consider a disk that has 10 data blocks starting from block 14 through 23. Let there be two files on the disk: F1 and F2. The directory structure lists that the first data blocks of F1 and F2 are respectively 22 and 16. Given the FAT table entries as below, what are the data blocks allotted to F1 and F2? **[10]**

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(14,18); (15,17); (16,23); (17,21); (18,20); (19,15); (20, −1); (21, −1); (22,19); 23,14).

In the above notation, (x, y) indicates that the value stored in table entry x points to data block y.

(b) A certain file system uses 8-KB disk blocks. The median file size is 4 KB. If all files were exactly 4 KB, what fraction of the disk space would be wasted? Do you think the wastage for a real file system will be higher than this number or lower than it? **[10]**

0.5

CLO 5 Appraise Elements of Input Output Techniques

Q4: Suppose that a system uses DMA for data transfer from disk controller to main memory. Further assume that it takes t_1 nsec on average to acquire the bus and t_2 nsec to transfer one word over the bus ($t_1 >> t_2$). After the CPU has programmed the DMA controller, how long will it take to transfer 1200 words from the disk controller to main memory, if (a) word-at-a-time mode is used, (b) burst mode is used? Assume that commanding the disk controller requires acquiring the bus to send one word and acknowledging a transfer also requires acquiring the bus to send one word. **[20]**

14 | 18

15 | 17

16

$\frac{4}{8} = 0.5$

20

$1-(0.5)^2 = 0.75$

$6 \times 2 + 10$

$1000(t_1+t_2) + 1000 t_2 + t_1 + t_2$

$t_1 + t_2 + 1000 t_2 + t_1 + t_2$

$2t + 1002 t_2$

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